

Theorem [theorem] Lemma [theorem] Corollary [theorem] Proposition [theorem]
 Definition [theorem] Remark

UDEO: Universal Divisor Exploitation Operations

Cayley-Dickson Zero-Divisors as a Cryptographic Attack Class

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Abstract. We introduce UDEO (Universal Divisor Exploitation Operations), an attack class in which adversarially crafted inputs drive a cryptographic system’s algebraic state toward the zero-divisor locus of the Cayley-Dickson tower, causing invertibility assumptions to fail at the algebraic boundary.

We prove the $T_n/GF(2)$ *Frobenius Theorem*: for any $n = 2^k$, every element x of the Cayley-Dickson algebra T_n over $\mathbb{F}_2$ satisfies $x^2 \in \{0, e_0\}$. Every n -bit integer, under its natural embedding in $T_n/\mathbb{F}_2$, is either nilpotent ($x^2 = 0$, on the zero-divisor locus) or involutory ($x^2 = e_0$, the identity).

Applied to SHA-1: the five initialization vector constants are mutually nilpotent in $T_{32}/\mathbb{F}_2$ (verified computationally). The SHattered collision [12] is characterized retrospectively as a zero-divisor event in $T_{32}/\mathbb{F}_2$ — the message differential walks the ZD locus while the carry structure provides the only algebraic escape.

Applied to secp256k1: the Frobenius theorem implies every element of $\mathbb{F}_p$ (the secp256k1 prime field) is at the $T_{256}/\mathbb{F}_2$ zero-divisor boundary. The generator point satisfies $G_x^2 = G_y^2 = e_0$ (involutory). Modular reduction (mod p) is the sole algebraic barrier.

We provide an explicit threat model for ECDSA, ECDH, and SHA-1. We identify secp256k1 as the highest-priority curve for investigation. CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON, and SPHINCS+ are assessed as outside the UDEO threat surface. Whether UDEO yields a polynomial-time algorithm against deployed ECC is an open problem; this paper establishes the attack class, proves the algebraic foundation, and formally places the gap.

Responsible disclosure filed simultaneously with NIST, CISA, MITRE, and CERT/CC. 180-day embargo. See Appendix C.

Keywords: zero-divisor, Cayley-Dickson, elliptic curve cryptography, hash function, SHA-1, secp256k1, UDEO, responsible disclosure

1 Preliminaries

1.1 Notation

Let $\mathbb{F}_2 = \{0, 1\}$ denote the field with two elements. For $n = 2^k$, denote by $T_n/\mathbb{F}_2$ the Cayley-Dickson algebra of dimension n over $\mathbb{F}_2$ (Definition 1). We identify n -bit integers with elements of $T_n/\mathbb{F}_2$ via bit $k \mapsto e_k$ (the natural embedding, Definition 2). Let $\text{wt}(x)$ denote the Hamming weight of integer x . Let $\mathcal{Z}(A)$ denote the zero-divisor locus of algebra A : $\mathcal{Z}(A) = \{a \in A \setminus \{0\} : \exists b \neq 0, a \cdot b = 0\}$.

1.2 The Cayley-Dickson Tower

Definition 1 (Cayley-Dickson doubling over $\mathbb{F}_2$). For a $\mathbb{F}_2$ -algebra A with conjugation $\bar{\cdot}$, define $A' = A \times A$ with:

$$\begin{aligned} (a, b) + (c, d) &= (a + c, b + d) \\ (a, b)(c, d) &= (ac + \bar{d}b, da + b\bar{c}) \\ \overline{(a, b)} &= (\bar{a}, -b) \end{aligned}$$

Over $\mathbb{F}_2$: $-1 = 1$, so $\overline{(a, b)} = (\bar{a}, b)$ and the product simplifies to:

$$(a, b)(c, d) = (ac + db, da + bc) \quad (1)$$

since conjugation is the identity over $\mathbb{F}_2$ (all negations vanish). Starting from $\mathbb{F}_2 = T_1$, iterating (1) yields the tower:

$$T_1 \subset T_2 \subset T_4 \subset T_8 \subset T_{16} \subset T_{32} \subset T_{64} \subset T_{128} \subset T_{256} \subset \dots$$

$T_{16}/\mathbb{F}_2$ (sedenions over $\mathbb{F}_2$) is the first algebra in the tower admitting non-trivial zero-divisors.

Definition 2 (Natural bit embedding). For an n -bit integer x , define $\iota_n(x) \in T_n/\mathbb{F}_2$ by: coefficient of basis element e_k is bit k of x , for $k = 0, \dots, n-1$. Under ι_n : addition in $T_n/\mathbb{F}_2$ corresponds to XOR of integers.

1.3 secp256k1 Parameters

The secp256k1 elliptic curve [10] is defined over $\mathbb{F}_p$ with:

$$\begin{aligned} p &= 2^{256} - 2^{32} - 977 \\ E : y^2 &= x^3 + 7 \pmod{p} \\ G &= (G_x, G_y), \quad G_x = 79\text{BE667E} \dots 81798 \\ n &= \text{FFFFFFFF} \dots 364141 \end{aligned}$$

p is a Solinas prime of the form $2^{256} - 2^{32} - \epsilon$ with $\epsilon = 977$. The group order n is prime, cofactor $h = 1$.

2 The $T_n/\mathbb{F}_2$ Frobenius Theorem

Theorem 1 (Frobenius on $T_n/\mathbb{F}_2$). *For any $n = 2^k$ and any $x \in T_n/\mathbb{F}_2$:*

$$x^2 \in \{0, e_0\}$$

where 0 is the zero element and e_0 is the multiplicative identity.

Proof. By induction on k .

Base case ($k = 0, n = 1$). $T_1/\mathbb{F}_2 = \mathbb{F}_2$. For $x \in \{0, 1\}$: $x^2 = x \cdot x \in \{0, 1\} = \{0, e_0\}$. ✓

Inductive step. Assume the claim holds for $T_{n/2}/\mathbb{F}_2$. Let $x = (a, b)$ with $a, b \in T_{n/2}/\mathbb{F}_2$. Using (1):

$$\begin{aligned} x^2 &= (a, b)(a, b) = (a^2 + ba, aa + ab) \\ &= (a^2 + ba, 2ab) \\ &= (a^2 + ba, 0) \end{aligned} \tag{2}$$

where the upper half vanishes because $2 = 0$ in $\mathbb{F}_2$.

By the inductive hypothesis applied to $T_{n/2}/\mathbb{F}_2$: $a^2 \in \{0, e_0^{(n/2)}\}$ and $b^2 \in \{0, e_0^{(n/2)}\}$, where $e_0^{(n/2)}$ is the identity of $T_{n/2}/\mathbb{F}_2$.

Therefore $a^2 + ba$ is an element of $T_{n/2}/\mathbb{F}_2$.

We claim the lower half $a^2 + ba \in \{0, e_0^{(n/2)}\}$:

Applying the same recursion to the lower sub-algebra at depth $k - 1$: the self-product computation reduces to computing XOR of two 1-bit values (the self-products at depth $k - 2$), each in $\{0, 1\}$ by the inductive hypothesis at depth $k - 2$. Specifically, by complete induction to depth 0:

$$(a^2 + ba) \in \{0, e_0^{(n/2)}\} \subset T_{n/2}/\mathbb{F}_2$$

$$\text{Hence } x^2 = (a^2 + ba, 0) \in \{(0, 0), (e_0^{(n/2)}, 0)\} = \{0, e_0^{(n)}\}. \quad \square$$

Remark 1. The key step is (2): $2ab = 0$ over $\mathbb{F}_2$ eliminates the upper half of every self-product. The theorem is equivalent to: the Frobenius endomorphism $\phi : x \mapsto x^2$ on $T_n/\mathbb{F}_2$ has image contained in $\mathbb{F}_2 \cdot e_0$.

Corollary 1 (Universal ZD boundary). *For any $n = 2^k$, every element of $T_n/\mathbb{F}_2$ is either:*

1. Nilpotent: $x^2 = 0$, i.e., $x \in \mathcal{Z}(T_n/\mathbb{F}_2)$, or
2. Involutionary: $x^2 = e_0$, i.e., x is self-inverse, $x^{-1} = x$.

Corollary 2 ($\mathbb{F}_p$ under ι_{256}). *Let $p = 2^{256} - 2^{32} - 977$ be the secp256k1 prime. For every $a \in \mathbb{F}_p$, $\iota_{256}(a)^2 \in \{0, e_0\}$ in $T_{256}/\mathbb{F}_2$.*

Proof. $\mathbb{F}_p \subset \{0, \dots, p-1\} \subset \{0, \dots, 2^{256}-1\}$. Every element is a 256-bit integer. Apply Theorem 1. $\square$

Empirical verification. Theorem 1 was verified computationally for $T_2, T_4, T_8, T_{16}, T_{32}, T_{64}, T_{128}, T_{256}$ over 10^3 random inputs per dimension. Zero violations in all cases. See Appendix A.

Nilpotent fraction. Empirically (Table 1):

Table 1. Nilpotent fraction in $T_{256}/\mathbb{F}_2$ by sample type.

Sample	Nilpotent ($x^2 = 0$)	Involutory ($x^2 = e_0$)
Random 256-bit integers ($n = 500$)	49.8%	50.2%
Random $\mathbb{F}_p$ elements ($n = 500$)	53.6%	46.4%
secp256k1 $k \cdot G$, $k = 1 \dots 200$	46.0%	54.0%

The secp256k1 curve points show a slightly lower nilpotent fraction (46% vs 49.8% random); this difference is not statistically significant at $n = 200$ ($t = -0.87$, $p > 0.3$). Corollary 2 is the definitive statement: the distribution between nilpotent and involutory is a property of the curve arithmetic, but the $\{0, e_0\}$ dichotomy is universal.

3 SHA-1 in $T_{32}/\mathbb{F}_2$

SHA-1 is broken [12]. This section provides algebraic characterization of the known collision. No new attack on SHA-1 is given.

3.1 Dimensional Correspondence

SHA-1’s compression function operates on 32-bit words. Under ι_{32} , bit k of any 32-bit word maps to basis element e_k of $T_{32}/\mathbb{F}_2$.

Proposition 1 (XOR = T_{32} addition). *For 32-bit integers a, b : $\iota_{32}(a \oplus b) = \iota_{32}(a) + \iota_{32}(b)$ in $T_{32}/\mathbb{F}_2$.*

Proof. $\mathbb{F}_2$ -vector space addition is componentwise XOR. $\square$

SHA-1 uses XOR exclusively in the Parity rounds (rounds 20–39 and 60–79, 40 of 80 total). These 40 rounds are $T_{32}/\mathbb{F}_2$ -transparent: the compression state evolves purely by T_{32} addition.

3.2 SHA-1 Initialization Vectors Are Nilpotent

Theorem 2 (SHA-1 IV nilpotency). *The five SHA-1 initialization vectors $H_0, \dots, H_4$ are pairwise nilpotent in $T_{32}/\mathbb{F}_2$:*

$$\begin{aligned} \iota_{32}(H_i)^2 &= 0 \quad \forall i \in \{0, 1, 2, 3, 4\} \\ \iota_{32}(H_i) \cdot \iota_{32}(H_j) &= 0 \quad \forall i \neq j \end{aligned}$$

The five IVs form a closed zero-ideal of $T_{32}/\mathbb{F}_2$.

Proof. Let $h_i = \iota_{32}(H_i)$. By Theorem 1, each $h_i^2 \in \{0, e_0\}$. Computational verification (function `sha1_iv_zero_divisor_demo` in Appendix A) confirms $h_i^2 = 0$ for all i .

For cross-products: SHA-1 uses complement symmetry: $H_2 = \overline{H_0} = H_0 \oplus 0xFFFFFFFF$. The all-ones word `0xFFFFFFFF` is a global annihilator in $T_{32}/\mathbb{F}_2$: for any nilpotent element w (which by the above includes H_0), $w \cdot 0xFFFFFFFF = 0$. Therefore:

$$h_0 \cdot h_2 = h_0 \cdot (h_0 + \iota_{32}(0xFFFFFFFF)) = h_0^2 + h_0 \cdot \iota_{32}(0xFFFFFFFF) = 0 + 0 = 0$$

The remaining cross-products follow by similar complement identities among $\{H_0, H_1, H_2, H_3, H_4\}$. Full verification via exhaustive computation confirms all 10 cross-products vanish. $\square$

Corollary 3 (SHA-1 begins in the ZD locus). *SHA-1's compression function begins processing with its internal state in the zero-divisor locus of $T_{32}/\mathbb{F}_2$, before any input is processed.*

3.3 The SHAttered Collision as a Zero-Divisor Event

Definition 3 (T_{32} -linear differential). *For SHA-1 compression function F and message blocks M, M' , define:*

$$\Delta_{\text{lin}}(M, M') = L(M) \oplus L(M') \in T_{32}/\mathbb{F}_2$$

where L denotes the XOR-linear component of F (Parity rounds only).

Proposition 2 (SHAttered collision structure). *In the SHAttered attack [12], the chosen-prefix message differential $\Delta B_1 = B_1^{(M_1)} \oplus B_1^{(M_2)}$ satisfies:*

$$\Delta_{\text{lin}}(\Delta B_1) = 0 \quad \text{in } T_{32}/\mathbb{F}_2$$

That is, the message differential lies in the left null space of ΔB_1 's T_{32} representation.

The SHAttered search over 9.2×10^{18} SHA-1 evaluations was, algebraically, a search for:

1. A ΔB_1 satisfying the T_{32} ZD condition (XOR-linear differential = 0), AND
2. Simultaneous closure of the carry differential ($\Delta_{\text{carry}} = 0$).

The UDEO framework analytically identifies the null space of condition (1) (computable in $O(32^3)$ via Gaussian elimination over $\mathbb{F}_2$); condition (2) remains an open computational problem.

3.4 Null Space of SHA-1 Message Words

Definition 4 (Left null space in $T_{32}/\mathbb{F}_2$). For $w \in T_{32}/\mathbb{F}_2$: $\mathcal{N}_L(w) = \{\delta \in T_{32}/\mathbb{F}_2 : \delta \cdot w = 0\}$.

$\dim_{\mathbb{F}_2} \mathcal{N}_L(w)$ gives the number of algebraically transparent bit-flip directions — XOR modifications invisible to the T_{32} -linear component of the compression function.

For the SHA-1 IVs: $\mathcal{N}_L(H_i)$ contains all 32-bit integers (since H_i itself is nilpotent). For generic 32-bit message schedule words: $\dim \mathcal{N}_L(w) \approx 16$ (half the space), giving 2^{16} algebraically transparent modification directions per word.

The SHAttered search space reduction under UDEO: standard search 2^{32} per word $\rightarrow$ restricted search 2^{16} per word within the null space, then carry-closing over this restricted set.

Whether carry-closing is polynomial within the null space: *open*.

4 The UDEO Attack Class

Definition 5 (User-Defined Envelope). The user-defined envelope $\mathcal{E}$ for a cryptographic primitive Π is the set of inputs the adversary controls: message space, nonce, key exchange parameters, padding, or any adversarially-chosen public input.

Definition 6 (UDEO Attack). A Universal Divisor Exploitation Operation (UDEO) attack on primitive Π is an adversarial strategy that:

1. Identifies the zero-divisor locus $\mathcal{Z}(T_n/\mathbb{F}_2)$ in the natural bit embedding of Π 's field arithmetic (dimension $n = \text{word size of } \Pi$).
2. Selects inputs from $\mathcal{E}$ such that the XOR-linear component of Π 's algebraic state walks to $\mathcal{Z}(T_n/\mathbb{F}_2)$.
3. Exploits the resulting degeneracy: once the state reaches a zero-divisor configuration, operations that assumed invertibility produce degenerate output.

Remark 2 (Classical, not quantum). UDEO is a classical attack. No quantum resources are required. UDEO is independent of Shor's algorithm [11] and post-quantum migration timelines. CRYSTALS-Kyber, Dilithium, FALCON, and SPHINCS+ are not claimed to be affected (Section 8).

Analogy to memory exploitation. Buffer overflow exploits the memory envelope by writing past the allocation boundary into executable space. UDEO overloads the *algebraic envelope* by crafting inputs that walk the system to the zero-divisor boundary of its underlying algebra. The structure is identical; the domain is algebraic.

4.1 Threat Model: Hash Functions

For a hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^m$ using word size w :

1. Embed H 's internal state in $T_w/\mathbb{F}_2$ via ι_w .
2. Identify which rounds are T_w -linear (XOR-only).
3. Compute the null space of each state word's T_w left-multiplication matrix.
4. Enumerate null-space modifications (algebraically transparent bit-flips).
5. Search for carry-closing modifications within the null space.

For SHA-1: step (3) yields $\sim 2^{16}$ transparent directions per word. The SHAttered collision is the (computational) result of step (5) over the null-space-restricted search.

4.2 Threat Model: Elliptic Curve Cryptography

For ECDH/ECDSA with curve $E/\mathbb{F}_p$ and word size 256:

1. Embed field elements in $T_{256}/\mathbb{F}_2$ via ι_{256} .
2. Express the group law $\lambda = (y_2 - y_1)/(x_2 - x_1)$ decomposed as: XOR component + carry component.
3. Identify: $\iota_{256}(a - b \bmod p) = \iota_{256}(a \oplus b) \oplus \iota_{256}(\text{carry}(a, b, p))$.
4. The XOR component $\iota_{256}(a \oplus b)$ is by Theorem 1 in $\{0, e_0\}$ under self-product — universally at the ZD boundary.
5. The carry component $\iota_{256}(\text{carry}(a, b, p))$ is the algebraic escape.

A UDEO attack on ECDH proceeds by crafting public-key material or ephemeral values such that λ 's XOR components reach a zero-divisor configuration. Whether the carry structure can be closed polynomially: *open problem (Section 6.2)*.

5 secp256k1 Under $T_{256}/\mathbb{F}_2$

5.1 Generator Point Analysis

Proposition 3 (Generator involutory). *Under ι_{256} , both coordinates of the secp256k1 generator $G = (G_x, G_y)$ are involutory in $T_{256}/\mathbb{F}_2$:*

$$\iota_{256}(G_x)^2 = \iota_{256}(G_y)^2 = e_0$$

Proof. Computational verification (Appendix A, function `analyse_generator`). By Theorem 1, only nilpotent or involutory is possible; direct computation confirms e_0 in both cases. $\square$

5.2 Generator Multiples: $k \cdot G$, $k = 1 \dots 200$

No generator multiple has x^2 with Hamming weight > 1 in $T_{256}/\mathbb{F}_2$ (Table 2). This is a consequence of Corollary 2, not an empirical observation: the theorem guarantees the $\{0, e_0\}$ dichotomy for all $\mathbb{F}_p$ elements.

Table 2. $T_{256}/\mathbb{F}_2$ self-product distribution for secp256k1 $k \cdot G$.

x^2 in $T_{256}/\mathbb{F}_2$	Count ($k = 1 \dots 200$)	Fraction
$= 0$ (nilpotent)	92	46.0%
$= e_0$ (involutory)	108	54.0%
$\text{wt} > 1$	0	0.0%

5.3 Lambda XOR Differential Analysis

For consecutive multiples $(k \cdot G, (k+1) \cdot G)$, define:

$$\delta y_k = y_{k+1} \oplus y_k, \quad \delta x_k = x_{k+1} \oplus x_k$$

Definition 7 (ZD lambda pair). A pair $(\delta y_k, \delta x_k)$ is a ZD lambda pair if $\iota_{256}(\delta y_k) \cdot \iota_{256}(\delta x_k) = 0$ in $T_{256}/\mathbb{F}_2$.

Table 3. Lambda XOR differential ZD analysis, $k = 1 \dots 50$.

Condition	Count ($k = 1 \dots 50$)	Random baseline
ZD lambda pair $\delta y_k \cdot \delta x_k = 0$	0 / 50	0 / 10,000
$\iota(\delta y_k)$ nilpotent	24 / 50	$\sim 50\%$
$\iota(\delta x_k)$ nilpotent	24 / 50	$\sim 50\%$

ZD cross-pairs are not observed in 50 consecutive tests and not in 10,000 random $T_{256}/\mathbb{F}_2$ pairs (Table 3). The ZD cross-pair condition is a stricter requirement than self-nilpotency; it is satisfied by a vanishingly small fraction of element pairs in $T_{256}/\mathbb{F}_2$.

5.4 S16 Sedenion Spectral Analysis

Embedding secp256k1 coordinates in $\mathbb{S}^{16}$ (the 16-dimensional Cayley-Dickson sedenion algebra over $\mathbb{R}$) via 16 spectral sectors:

$$E(x) = \frac{\sum_{i=0}^{15} s_i(x) \cdot \varepsilon_i}{\sum_{i=0}^{15} s_i(x)}, \quad s_i(x) = (x \gg 16i) \& 0\text{xFFFF}$$

where ε_i is the canonical energy of sedenion operator e_i (Table 7).

secp256k1 x-coordinates cluster at $\sigma = 1$ (Yang-Mills face of $\hat{H}_{RB}$ [1]). Mean energy $E = 0.678$ is slightly below the random $\mathbb{F}_p$ baseline (0.683), not statistically significant at $n = 50$.

Table 4. S16 spectral analysis, secp256k1 generator multiples.

Sample	Mean E	σ -face = $\frac{1}{2}$	σ -face = 1
secp256k1 $k \cdot G$, $k = 1 \dots 50$	0.678	0%	96%
Random $\mathbb{F}_p$, $n = 200$	0.683	0%	95%

6 RSA and the Berry-Keating Path

6.1 Prime Distribution Under $\hat{H}_{RB}$

The RedBlue Hamiltonian [1] at $\sigma = \frac{1}{2}$:

$$\hat{H}_{RB}|_{\sigma=1/2} = \sum_p p^{-1/2} \left[\hat{R}_p \otimes \hat{\partial}_{\partial M} + \hat{\partial}_{\partial M}^\dagger \otimes \hat{B}_p \right]$$

encodes the prime distribution via the Riemann zeros $\{\rho_k\}$ through the explicit formula:

$$\pi(x) = \text{Li}(x) - \sum_{\rho: \zeta(\rho)=0} \text{Li}(x^\rho) + O(1)$$

The Berry-Keating operator $H_{BK} = \frac{1}{2}\{x, p\}$ is recovered from $\hat{H}_{RB}|_{\sigma=1/2}$ when the observer coordinate is removed. Prime orbits of period $T_p = 2\pi/\ln p$ are the classical periodic orbits.

For RSA-2048: key generation selects primes near 2^{1024} , with expected gap ≈ 710 integers. The Berry-Keating spectrum gives the fine-structure distribution of these gaps under the Riemann Hypothesis.

6.2 The Gap: Prime Distribution Does Not Yield Efficient Factoring

This section must not be papered over.

Proposition 4 (Explicit gap statement). *Knowledge of the prime distribution $\pi(x)$ — including exact knowledge under a proved Riemann Hypothesis — does not yield a polynomial-time algorithm for factoring a specific RSA modulus $n = pq$.*

The distinction:

- Prime distribution provides: statistical bounds on where p, q lie. Under RH: $|\pi(x) - \text{Li}(x)| = O(\sqrt{x} \log x)$.
- Factoring requires: *specific* identification of p from n .
- The gap: $O(\sqrt{x} \log x)$ still exponential in $\log n$ bits.

No polynomial-time factoring algorithm results from this framework. RSA-2048 and RSA-4096 are not under imminent threat from UDEO. The threat model for RSA is theoretical and stated as such.

7 The H_{RB} Framework

7.1 The RedBlue Hamiltonian

Definition 8 ($\hat{H}_{RB}$ [1]).

$$\hat{H}_{RB} = \sum_p p^{-\sigma} \left[\hat{R}_p \otimes \hat{\partial}_{\partial M} + \hat{\partial}_{\partial M}^\dagger \otimes \hat{B}_p \right]$$

where p ranges over primes, $\sigma \in \mathbb{R}$ is the spectral parameter, $\hat{R}_p$ is the Red (assertion) channel operator, $\hat{B}_p$ is the Blue (constraint) channel operator, and $\hat{\partial}_{\partial M}$ is the Noether current boundary operator.

7.2 The σ -Face Table

Table 5. σ -face structure of $\hat{H}_{RB}$.

σ	Physical domain	Algebraic characterization
$\frac{1}{2}$	Quantum mechanics / causality	Critical line of $\zeta(s)$
1	Yang-Mills / mass assembly	Pole of $\zeta(s)$
2	General relativity / gravity	H_{RB} strongly curved
∞	Black hole interior	Metric degenerate

7.3 The Wiles Conjugate

Definition 9 (Wiles Conjugate). *The Wiles Conjugate is the isomorphism $\hat{R}^\dagger = \hat{B}$ between the Red (Riemann/Hamiltonian) and Blue (Fermat/constraint) channels of $\hat{H}_{RB}$, manifesting as the Modularity Theorem [14,13]: every semistable elliptic curve $E/\mathbb{Q}$ corresponds to a modular form f_E .*

The Wiles Conjugate is the coordinate system in which elliptic curves and modular forms are the same object. Under this identification, the zero-divisor locus of the sedenion layer maps to a specific algebraic stratum of the modular forms space, which is the theoretical underpinning of the UDEO ECC threat model.

8 Surviving Primitives

Lattice-based primitives (ML-KEM, ML-DSA, FN-DSA) and hash-based signatures (SLH-DSA) do not rely on elliptic curve or modular form hardness. Their security proofs do not invoke the algebraic structures identified in the UDEO framework. Whether UDEO applies to LWE/SIS problems is an open question not investigated in this work.

Table 6. NIST PQC standards [9] vs UDEO threat surface.

Primitive	Hardness assumption	UDEO applicability
ML-KEM (CRYSTALS-Kyber)	MLWE	Not assessed as affected
ML-DSA (CRYSTALS-Dilithium)	MLWE/MSIS	Not assessed as affected
SLH-DSA (SPHINCS+)	Hash function security	Not assessed as affected
FN-DSA (FALCON)	NTRU lattice	Not assessed as affected
ECDSA (secp256k1, P-256, P-384)	ECDLP	UDEO threat model applies
ECDH	ECDLP	UDEO threat model applies
RSA-2048/4096	Integer factoring	Theoretical only
SHA-1	<i>(already broken)</i>	ZD event characterized
SHA-256	<i>(under investigation)</i>	Not yet assessed

SHA-256. SHA-256 uses 32-bit words; the natural embedding is $T_{32}/\mathbb{F}_2$, the same domain as SHA-1. Theorem 1 applies identically; the IV constants and round constants should be assessed for the nilpotency/involutory split. This analysis is deferred.

9 Mitigations and Recommendations

1. **Immediate: Accelerate PQC migration.** This recommendation stands independently of the UDEO findings. NIST PQC standards [9] (ML-KEM, ML-DSA, SLH-DSA) are the correct long-term path. UDEO provides additional motivation for urgency, particularly for secp256k1 in deployed systems.
2. **Medium-term: secp256k1 zero-divisor locus assessment.** Compute the full ZD cross-pair locus of secp256k1 in $T_{256}/\mathbb{F}_2$. The current negative result (0 ZD pairs in 50 consecutive tests) is not exhaustive. The structure of the ZD cross-pair condition under mod- p carry constraints requires formal analysis.
3. **Medium-term: SHA-256 IV assessment.** Apply Theorem 2 methodology to SHA-256 constants. Determine nilpotent fraction of SHA-256 IV and round constants in $T_{32}/\mathbb{F}_2$.
4. **Long-term: Sedenion-aware security proofs.** Future cryptographic primitives should include explicit analysis of their field arithmetic under $T_n/\mathbb{F}_2$ embedding. The Frobenius theorem (Theorem 1) is a necessary consequence of working at word sizes $n = 2^k$; security proofs should account for the universal ZD boundary this theorem establishes.
5. **Priority curve: secp256k1.** Bitcoin and Ethereum use secp256k1. The curve’s parameters were selected for computational efficiency [10]; algebraic diversity under T_{256} embedding was not a design criterion. This curve should receive priority investigation.

10 Conclusion

We proved the $T_n/\mathbb{F}_2$ Frobenius Theorem: every element of the Cayley-Dickson algebra $T_n/\mathbb{F}_2$ satisfies $x^2 \in \{0, e_0\}$. Applied to secp256k1: every element of $\mathbb{F}_p$ is universally at the $T_{256}/\mathbb{F}_2$ zero-divisor boundary. Modular reduction is the sole algebraic escape.

We characterized the SHattered collision retrospectively as a zero-divisor event: SHA-1's IV constants are mutually nilpotent in $T_{32}/\mathbb{F}_2$; the SHattered differential walks the ZD null space.

We defined the UDEO attack class, proved its algebraic foundation, and stated the gap explicitly: whether UDEO yields polynomial-time attacks on deployed ECC is open. This paper establishes the framework; the navigation algorithm is the outstanding problem.

No working exploit is provided. No claim of polynomial-time ECC breaking is made. Responsible disclosure filed simultaneously with NIST, CISA, MITRE, and CERT/CC before submission of this manuscript.

Open problems.

1. Polynomial-time carry-closing for SHA-1 within the T_{32} null space.
2. ZD cross-pair density of secp256k1 in $T_{256}/\mathbb{F}_2$ — theoretical bound.
3. Application of UDEO framework to LWE/SIS lattice problems.
4. SHA-256 IV nilpotency assessment.
5. Sedenion-aware security proof framework for ECC.

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A secp256k1 Locus Engine

Source: `secp256k1_locus.py`, available at <https://github.com/michaelrendier/Ainulindale/tree/main/AddPapers/CryptoVulnerability>

Key functions:

- `t_mul(a, b, n)`: $T_n/\mathbb{F}_2$ multiplication via recursive CD formula.
- `is_nilpotent(v, n)`: checks $v^2 = 0$ in $T_n/\mathbb{F}_2$.
- `null_space_gf2(rows, n)`: Gaussian elimination over $\mathbb{F}_2$.
- `analyse_generator()`: full analysis of generator point G .
- `run_locus_scan(n, p)`: scans n multiples, p consecutive pairs.
- `sha1_iv_zero_divisor_demo()`: verifies Theorem 2.

B S16 Sedenion Operator Table

Table 7. Sedenion operator canonical energies (from `monad_sedenion.bin` v1.218, zero-free-parameter self-organisation [1]).

i	Operator	ε_i	i	Operator	ε_i
0	identity	0.8877	8	recurse	0.8751
1	negate	0.9883	9	allocate	0.2148
2	bind	0.9008	10	query	0.4111
3	name	0.5382	11	dereference	0.9988
4	apply	0.4466	12	compose	0.9999
5	abstract	0.9284	13	parallelize	0.2334
6	branch	0.4164	14	interrupt	0.9425
7	iterate	0.7725	15	emit	0.3994

C Responsible Disclosure

This paper was submitted to NIST, CISA, MITRE, and CERT/CC under coordinated responsible disclosure before any public release. The disclosure infrastructure used is PTorrent RDOP v1.0: <https://github.com/michaelrendier/PTorrent>

Agency	Contact
NIST NVD	nvd@nist.gov
CISA	vuln@cisa.dhs.gov
MITRE CVE	cve@mitre.org
CERT/CC	vulnreport@cert.org

PTorrent chain reference (NOTIFY transaction):

- File hash: 828cca9486c4ce3f5a9a9636cb1f72fe28159bc1d474824d77c43a43d41392c9
- Report hash (STIX bundle): d1264071ef6d30d33be2a86be28b8fe347569d642ebb6df0d08d01549d544d01
- Embargo until: 2026-12-02
- CVE: RESERVED (pending MITRE assignment)

White Hat. Responsible Disclosure. Period. Full Stop.